

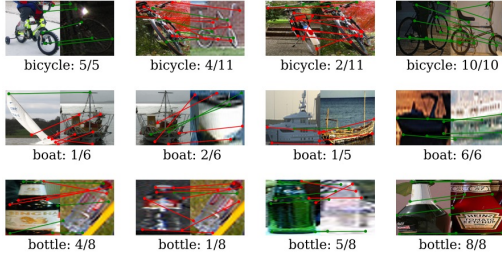
Self-supervised Learning of Visual Graph Matching

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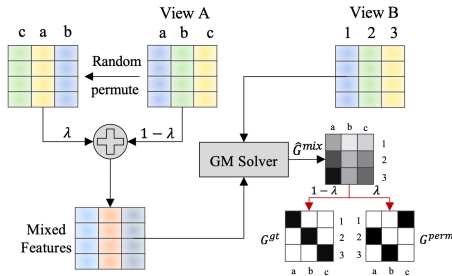
Self-supervised graph matching



Key points matching in visual graphs data

Two-stage data augmentation methods

1. Data augmentation on images (crop, flip, etc)
2. Data augmentation on graphs (Mix)



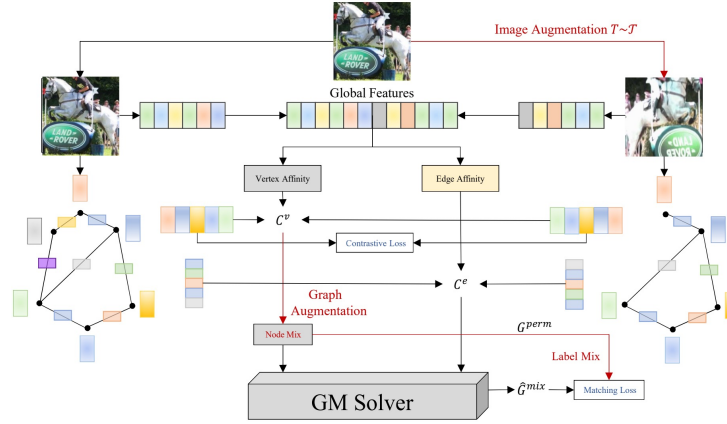
Node-level contrastive objective

$$\mathcal{L}_{node}^A = -\mathbb{E}_{ihw} \log \frac{\exp(\mathbf{C}_{ihw}^A \cdot \mathbf{C}_{ihw}^B)}{\exp(\mathbf{C}_{ihw}^A \cdot \mathbf{C}_{ihw}^B) + \mathcal{N}}$$

$$\text{, where } \mathcal{N} = \sum_{i'h'w'} \exp(\mathbf{C}_{ihw}^A \cdot \mathbf{C}_{i'h'w'}^B) + \exp(\mathbf{C}_{ihw}^A \cdot \mathbf{C}_{i'h'w'}^A)$$

$\mathcal{N}$ is the negative pairs in contrastive learning

Framework



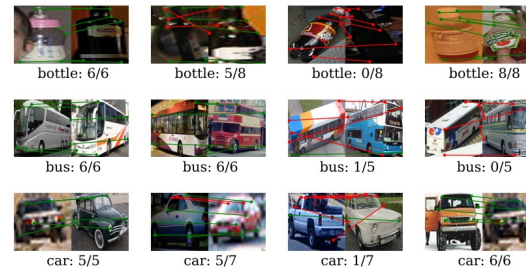
Framework of the proposed **SCGM** (Self-supervised Contrastive Graph Matching framework). The proposed two-stage augmentation modules are highlighted in red.

Node mix and label mix

$$\text{Node mix } \mathbf{x}_i = (1 - \lambda) \cdot \mathbf{x}_i + \lambda \cdot \mathbf{x}_j$$

$$\text{Label mix } \mathcal{L}_{match} = (1 - \lambda) \cdot \mathcal{L}(\hat{\mathbf{G}}^{mix}, \mathbf{G}^{gt}) + \lambda \cdot \mathcal{L}(\hat{\mathbf{G}}^{mix}, \mathbf{G}^{perm})$$

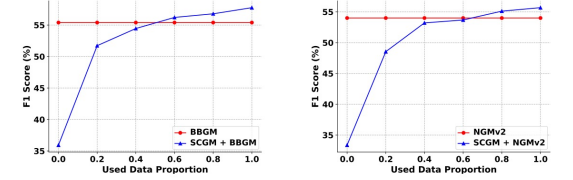
Visual matching results



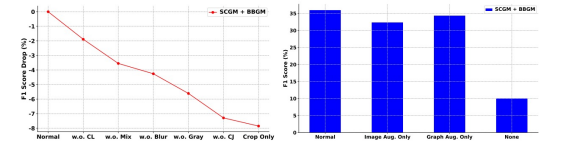
Experiments

Method	Label	airplane	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	motor	person	plant	sheep	sofa	train	tv	Avg
GMN [34]	✓	31.9	47.2	31.9	40.8	68.7	72.2	53.6	52.8	34.6	48.6	72.3	47.7	54.8	33.0	38.6	75.1	49.3	45.0	83.0	54.3
PCA [37]	✓	49.8	61.9	65.3	57.2	78.8	75.6	64.7	69.7	41.6	63.4	50.7	67.1	66.7	61.6	44.5	81.2	67.8	59.2	78.5	64.8
NGM [39]	✓	50.1	63.5	57.9	53.4	79.8	77.1	73.6	68.2	41.1	66.4	40.8	60.3	61.9	63.5	45.6	77.1	69.3	65.5	79.2	64.1
IPCA [37]	✓	53.8	66.2	67.1	61.2	80.4	75.3	72.6	72.5	44.6	65.2	54.3	67.2	67.9	64.2	47.9	84.4	70.8	64.0	83.8	67.7
CIE [40]	✓	52.5	66.6	70.2	57.1	82.1	77.0	70.7	73.1	43.8	69.9	62.4	70.3	70.3	66.4	47.6	85.3	71.7	64.0	83.8	68.9
BBGM [31]	✓	61.9	71.1	79.7	79.0	87.4	84.0	89.5	80.2	56.8	79.1	64.6	78.9	76.2	75.1	65.2	88.2	77.3	77.0	94.9	80.9
NGMv2 [31]	✓	61.8	71.2	77.6	78.8	87.3	83.6	87.7	79.8	55.4	77.8	89.5	78.8	80.1	79.2	62.6	97.7	77.7	75.7	96.7	80.1
IPFP [24]	✗	25.1	26.4	41.4	50.3	43.0	32.9	37.3	32.5	33.6	28.2	26.9	26.1	29.9	32.0	28.8	62.9	28.2	45.0	69.3	33.8
HRWM [41]	✗	30.9	40.0	46.4	54.1	52.3	35.6	47.4	37.3	36.3	34.1	28.8	35.0	39.1	36.2	39.5	67.8	38.6	48.4	70.5	41.3
PSM [6]	✗	32.6	37.5	40.9	53.2	47.8	34.6	50.1	35.5	37.2	36.3	28.1	32.7	42.4	37.1	36.5	62.9	43.7	54.3	72.6	40.8
GNCCP [25]	✗	28.9	37.1	46.2	53.1	48.0	35.3	45.5	34.7	36.3	34.2	25.2	35.3	39.8	39.6	40.7	61.9	37.4	50.5	67.0	34.8
BPF [41]	✗	30.9	40.4	47.3	54.5	50.8	35.1	46.7	36.3	40.9	38.9	16.3	34.8	39.8	39.6	39.3	63.2	37.9	50.2	70.5	41.3
GANN [36]	✗	19.2	20.5	24.1	27.9	30.8	50.9	36.4	22.3	24.4	23.2	39.8	21.7	20.5	23.9	15.8	42.2	29.8	17.1	61.8	31.5
SCGM (two-graph)	✗	37.6	49.9	54.8	54.5	65.6	56.4	60.6	52.3	36.8	51.4	50.4	47.2	59.4	51.2	38.3	91.3	59.3	52.7	83.1	57.1
SCGM + BBGM	✗	34.3	48.2	51.0	52.2	63.3	56.0	62.0	50.1	38.5	49.9	39.9	46.2	54.8	52.1	37.4	82.3	56.8	51.4	80.2	78.8

Average performance across all the objects w.r.t accuracy (%) on Pascal VOC with standard intersection filtering



Results after finetuning with part of the data in the Pascal VOC dataset for SCGM



F1 score drop by removing components from vanilla SCGM with BBGM as the backbone solver.

References

- [1] Wang R, Yan J, Yang X. Neural graph matching network: Learning lawler's quadratic assignment problem with extension to hypergraph and multiple-graph matching[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021.
- [2] Rolínek M, Swoboda P, Zietlow D, et al. Deep graph matching via blackbox differentiation of combinatorial solvers[C]//European Conference on Computer Vision. Springer, Cham, 2020: 407-424.